

EXPERT GAMES after 20.10.2024

test #	games	won	lost	% won	average time per game (s)	total time (s)	situations solved by minecount	minecount solutions / game	guesses	guesses / game	pic in dropbox
1	10006	3883	6123	38,81 %	0,087	871					"after mc bug fixes 10006 games won 38.81% time 871 ms per game"
2	10000	3985	6015	39,85 %	0,193	1930					"10000 games 39.9% won 1930 ms per game"
3	100002	38693	61309	38,69 %	0,148	14800					"100 002 games 38.7% won 14800 ms per game"
4	10013	3923	6090	39,18 %	0,198	1983					"expert 10 013 games 39.2% won 1983 ms per game"
5	11333	4348	6985	38,37 %	0,144	1632	2076	0,18			"11 333 expert games WITH MINECOUNT 38.37% won 1632 ms per game"
6	11119	4444	6675	39,97 %	0,143	1590	2103	0,189	29485	2,65	"11 119 expert games 40.0% won 1590 ms per game"
For test #7 onwards, (1) I had fixed a probability calculation error regarding front cells. After this fix, more front cell guesses were done on average than unclicked unseen cell guesses, unlike before (2) Faster thanks to self.variables bug fix, resulting in successful early return from minecount filtering situations if no variables can be solved											"expert 12 204 games 56 ms per game 38.6% after fixes - faster and just as good win percent"
7	12204	4705	7499	38,55 %	0,056	683	2208	0,181	31811	2,61	
utilizing average uu mine density in guessing											
8	8806	3424	5382	38,88 %	0,061	537	1587	0,18	23333	2,65	
utilizing max uu mine density in guessing											
9	10213	3956	6257	38,73 %	0,068	694	1790	0,175	27207	2,66	
going back to utilizing average uu mine density											
10	15484	6022	9462	38,89 %	0,061	945	2700	0,174	40678	2,63	"expert 15 484 games won 38.9% time 61 ms per game AFTER ALL FIXES UTILIZING AVERAGE UU MINE DENSITY"
based on the two blues (average uu mine density), best solving percent is using that (the average)											
TOTAL	183696	71361	112335	38,85 %	0,135	24720	9764	53675	111836	2,6	

utilizing average mine density in guessing and early no solutions -minecount detection

8	8806	3424	5382	38,88 %	0,061	537	1587	0,18	23333	2,65
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10	15484	6022	9462	38,89 %	0,061	945	2700	0,174	40678	2,63	"expert 15 845 games won 38.9% time 61 ms
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TO TA L	46359	17973	28386	38,77 %	0,068	3159	6987	46359	104689	0,4	
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added second condition for useless minecount, fixed timer-derived do nothing -loop

test #	games	won	lost	% won	average time per game (s)	total time (s)	situations solved by minecount	minecount solutions / game	guesses	guesses / game	pic in dropdown
1	10000	3906	6094	39,06 %	0,079	790	1969	0,197	26427	2,64	"after mc bug fixes 10006 gan
2	22069	8527	13542	38,64 %	0,076	1677	4225	0,191	59109	2,68	"expert test 2 25.10.2024"
3	18492	7203	11289	38,95 %	0,076	1405	3463	0,187	49685	2,69	"expert test 3 25.10.2024"
TAL	50561	19636	30925	38,84 %	0,077	3873	9657	0,191	135221	2,67	

BEGINNER GAMES

test #	games	won	lost	% won	average time per game (s)	total time (s)	situations solved by minecount	minecount solutions / game	guesses	guesses / game	pic in dropdown
1	25038	24006	1032	95,88 %	0,005	125					"beginner 95.9% won 25000 g
below: same setup as in expert test #8 and #10											
2	10027	9552	475	95,26 %	0,004	40	510	0,051	2479	0,25	

TO TA L	35065	33558	1507	95,70 %	0,004	125	510	10027	0	0	
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utilizing average mine density in guessing and early no solutions -minecount detection

2	10027	9552	475	95,26 %	0,004	40	510	0,051	2479	0,25	
TO TA L	10027	9552	475	95,26 %	0,004	40	510	10027	2479	0,1	

added the second condition for useless minecount, fixed timer-derived do nothing -loop

tes t #	games	won	lost	% won	avera ge time per game (s)	total time (s)	situa tions solve d by mine coun t	minec ount soluti ons / game	guesse s	gues ses /ga me	
1	23474	22462	1012	95,69 %	0,006	140,8	1282	0,055	5865	0,25	
2											
TO TA L	23474	22462	1012	95,69 %	0,006	140,8	1282	0,055	5865	0,25	

INTERMEDIATE GAMES

tes t #	games	won	lost	% won	avera ge time per game (s)	total time (s)	situa tions solve d by mine coun t	minec ount soluti ons / game	guesse s	gues ses /ga me	
1	26430	22085	4345	83,56 %	0,011	291					
2	13170	11063	2107	84,00 %	0,018	237	1477	0,112	9524	0,72	
below: same setup as in expert test #8 and #10											
3	13357	11051	2306	82,74 %	0,012	160	1425	0,107	9807	0,73	
TO TA L	39600	33148	6452	83,71 %	0,013	528	1477	13170	9524	0,7	

[pic in dropbox](#)

"Int 26430 games 83.6% won

utilizing average mine density in guessing and early no solutions -minecount detection

3	13357	11051	2306	82,74 %	0,012	160	1425	0,107	9807	0,73	
TO TA L	13357	11051	2306	82,74 %	0,012	160	1425	13357	9807	0,1	

added the second condition for useless minecount, fixed timer-derived do nothing -loop

test #	games	won	lost	% won	average time per game (s)	total time (s)	situations solved by minecount	minecount solutions / game	guesses	guesses / game	
1	20145	16761	3384	83,20 %	0,012	241,7	2252	0,112	14651	0,73	"25.10 int test #1"
2											
TO TA L	20145	16761	3384	83,20 %	0,012	241,7	2252	0,112	14651	0,73	

The time per game does not include times in-between games (between game end and start) which includes pygame drawing the screen, and resetting variables for the game.

Solving: (1) simple_solver (2) old CSP (3) new CSP: 3.1 normal, 3.2

All tests had a 20-s timer limit for 'join_compatibility_groups...()' (beginning of CSP_solver). If this timer expires, a guess is issued. This timer rarely expires, and only does so in expert games.

dropbox link:

<https://www.dropbox.com/scl/fo/4dt7sda6htrzk6hp844tk/AOeWAB4MPLSIXZsdFEsb4tE?rlkey=e6065s75yqxx3f3ehtaypql4b&st=6lwjbten&dl=>

in GitHub, this link is in '/Testing/link_to_screenshots_and_videos'

3 ms per game (although I had 10 s sleep for manual checks which increases that) 20s timer"

